



Time = 30 Minutes

## Quiz No 9

[20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

## Question 1 [20 points] Cumulative Distribution Function for Mixed Distribution

- **Discrete Random Variable X** - You toss a biased coin two times, where the RV  $X$  is the number Heads in the two throws. Therefore,  $X$  is a Binomial Random Variable with parameter  $n = 2$  with probability of success (Heads) =  $1/3$ .
- **Continuous Random Variable Y** - RV  $Y$  has a uniform distribution with  $0 \leq y \leq 3$ .
- **Random Variable Z** =  $X$  with probability 0.5 and  $Z = Y$  with probability 0.5. Random Variable  $Z$  has a mixed distribution.

(a) [8 points] Calculate and plot the Cumulative Distribution Function (CDF) for RV  $Z$ . Use the diagram given below (Figure 1) to plot the CDF, clearly marking the exact CDF heights.

(b) [4 points] Find the probability that RV  $Z$  takes on values in the interval,  $1 \leq z \leq 2$ .

(c) [6 points] Let's now assume that the Continuous Random Variable  $Y$  is a standard Gaussian RV with 0 mean and unit variance, i.e.  $Y \sim N(0, 1)$ . Again, Random Variable  $Z = X$  with probability 0.5 and  $Z = Y$  with probability 0.5. Calculate and plot the Cumulative Distribution Function (CDF) for RV  $Z$ . Use the diagram given below (Figure 2, Page 2) to plot the CDF, clearly marking the CDF heights (be precise as much as you can).

(d) [2 points] Find the probability (for the revised RV  $Y$  as in Part c) that the RV  $Z$  takes on values in the interval,  $-\infty \leq z < 0$ .

$$P_X(0) = \binom{2}{0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^2 = \frac{4}{9}, \quad P_X(1) = 2 \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{9}$$

$$P_X(2) = \frac{1}{9}$$

$$F_X(0) = \frac{4}{9}, \quad F_X(1) = \frac{8}{9}, \quad F_X(2) = 1$$

$$F_Z(z) = \frac{1}{2} \cdot F_X(z) + \frac{1}{2} F_Y(y)$$

$$F_Z(0) = \frac{1}{2} \cdot \frac{4}{9} = \frac{2}{9} = 0.222$$

$$F_Z(1) = \frac{1}{2} \cdot \frac{8}{9} + \frac{1}{2} \cdot \frac{1}{3} = 0.61$$

$$F_Z(2) = \frac{1}{2} + \frac{1}{2} \cdot \frac{2}{3} = 0.833$$

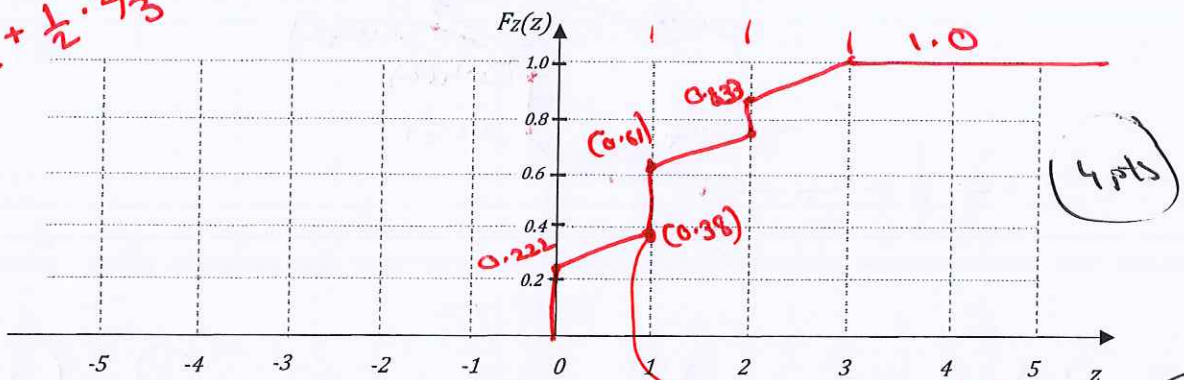
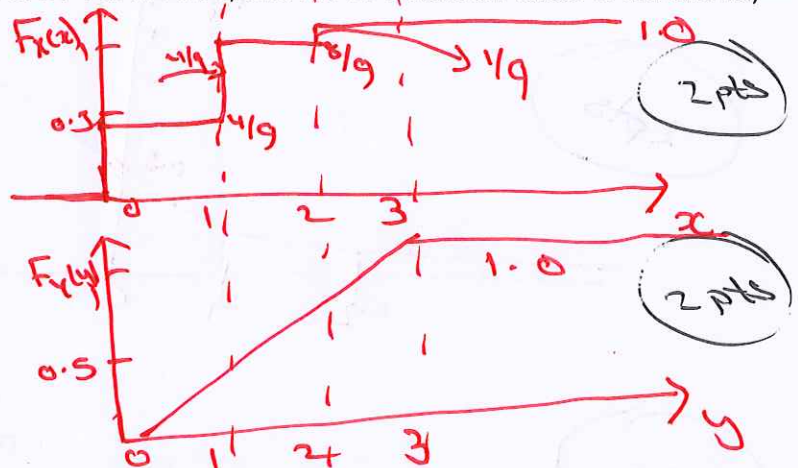


Fig.1 (for part a)

$$F_Z(3) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 1$$

(b)  $P(1 \leq Z \leq 2)$

$$= F_Z(2) - F_Z(1)$$

$$= 0.833 - 0.38$$

$$= 0.453$$

(c)  $F_X(0) = 4/9$ ,  $F_X(1) = 8/9$  &  $F_X(2) = 1.0$

$F_Y(-2) = 0.0228$ ,  $F_Y(-1) = 0.1587$ ,  $F_Y(0) = 0.5$ ,  $F_Y(1) = 0.8413$   
 $F_Y(2) = 0.9772$

$F_Z(z) = \frac{1}{2} \cdot F_X(x) + \frac{1}{2} F_Y(y)$

$F_Y(3) \approx 1$

$F_Z(-2) = 0.0114$ ,  $F_Z(-1) = 0.07935$ ,  $F_Z(0) = 0.4722$

$F_Z(1) = 0.865$ ,  $F_Z(2) = 0.9886$ ,  $F_Z(3) = 1$

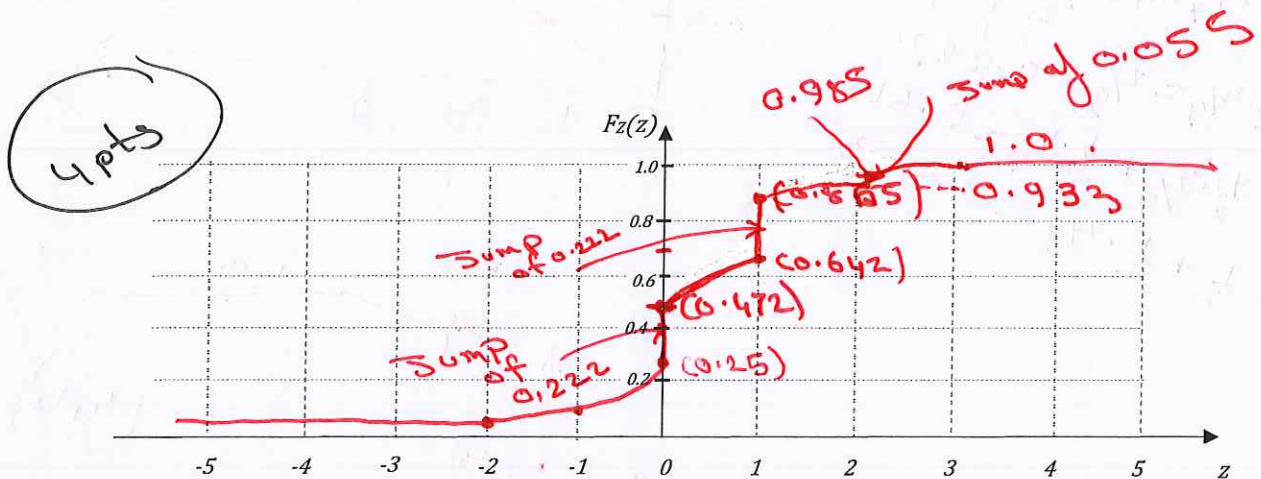
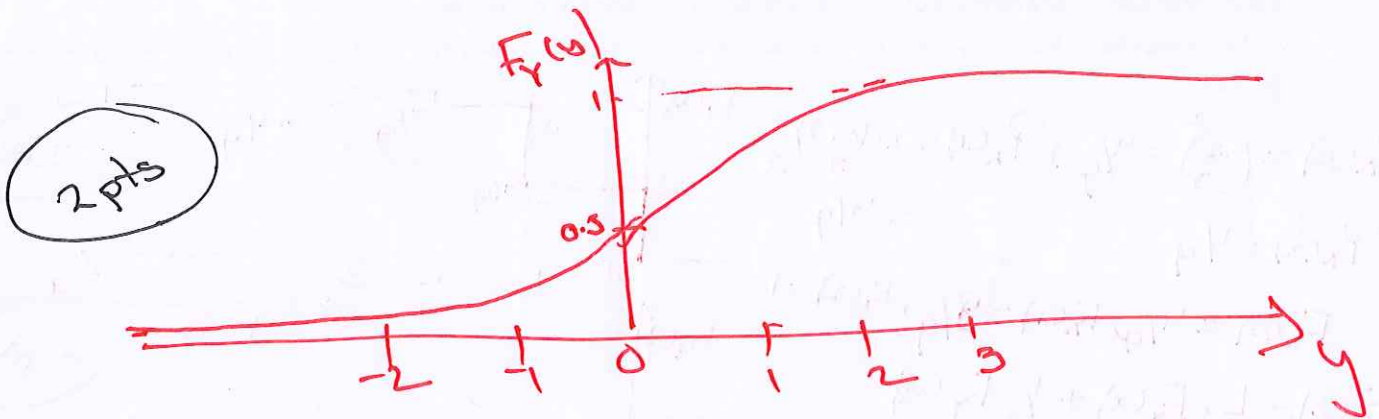


Fig. 2 (for part c)

(d)  $F_Z(0^-) - F_Z(-\infty) = P(-\infty \leq Z < 0)$   
 $= 0.25$

2 pts